K-means always finds groups, even when there are none

K-means sorts data points into k groups, and you choose k. Ask for four groups, and you almost always get four; a group can, rarely, end up empty (Harris, 2014).

How it works

K-means repeats two steps. Each point joins its nearest centre, then each centre moves to the average position of its points (University of Essex Online, 2026). The total squared distance to the centres (SSE) cannot go up, and there are only so many ways to group the points, so the loop always stops (Harris, 2014).

Animation 1: the start matters

Shabalin (no date) starts all four centres at the leftmost points. It ends with one group split and two merged. Of his five runs, only two find the four groups. Géron (2025) calls this a local optimum: a stable answer that is not the best one. The usual fix is several runs from different starts, keeping the lowest SSE.

Animation 2: groups that are not there

Harris's 'I'll Choose' with 'Uniform Points' spreads the points evenly, so there are no real groups. I bunched ten centres in the middle, and k-means still spread them into ten tidy coloured regions. Restarts cannot fix this. Two checks give clues, though neither is proof (University of Essex Online, 2026). The elbow plot shows where extra groups stop lowering the SSE much. The silhouette score asks whether each point sits closer to its own group than to the next; near 1 means clear groups (Géron, 2025). My Python test on 250 uniform points seemed to show an elbow at four, yet the silhouette stayed around 0.4 for every k tested. Four simulated groups scored about 0.77. My notebook is on GitHub: https://github.com/Tanky0/Tanky0.github.io/blob/main/assets/documents/ML_Unit05_KMeans_Check_Notebook.ipynb

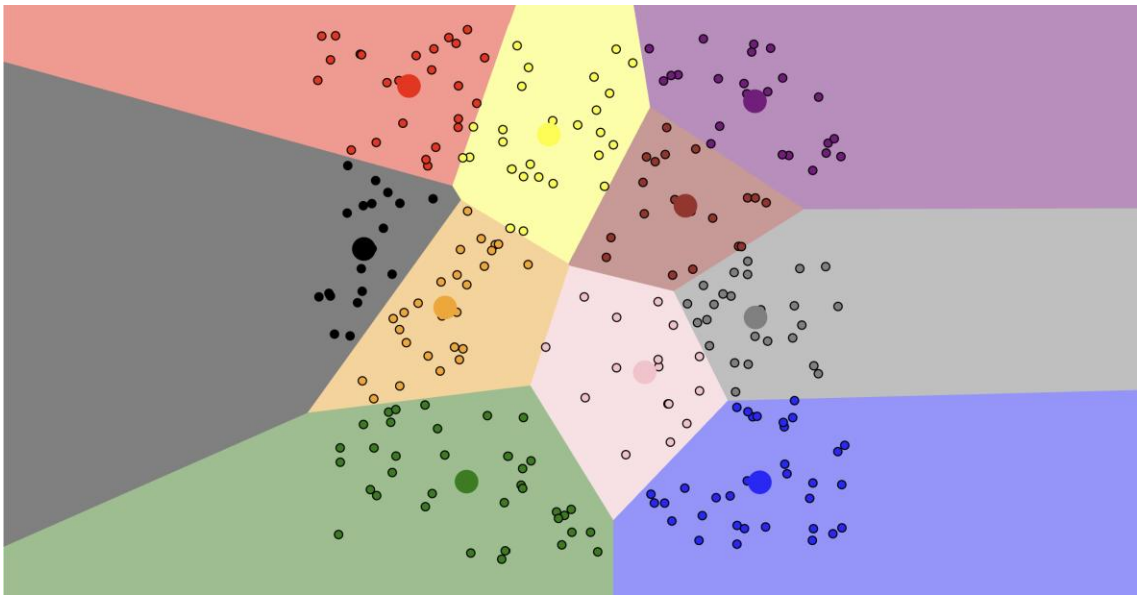


Figure 1: My run of Harris's (2014) animation with 'I'll Choose' and 'Uniform Points'.

Why it matters

Clusters are a useful first look, but not proof of real types. The lecturecast's insurer makes one offer per customer group (University of Essex Online, 2026); if the groups are artefacts, near-identical customers on either side of an invented boundary could get different deals. In 114,231 Swedish patients newly diagnosed with type 2 diabetes, simple clinical measures predicted death better than k-means clusters (Lugner et al., 2021).

Where in your own field would you accept a cluster as a real category?

References

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Lugner, M. et al. (2021) 'Comparison between data-driven clusters and models based on clinical features to predict outcomes in type 2 diabetes: nationwide observational study', *Diabetologia*, 64(9), pp. 1973–1981. Available at: https://doi.org/10.1007/s00125-021-05485-5

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